Advay Vyas

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Education

The University of Texas at Austin, Austin, TX — August 2024 – May 2027

Bachelors of Science in Statistics and Data Science & Mathematics, GPA 3.95

Relevant Coursework: Real Analysis I & II, Probability, Stochastic Processes, Mathematical Statistics, Linear Algebra, Differential Equations, Software Design, Databases, Multivariable Calculus.

Research Experience

Department of Statistics, The University of Texas at Austin, Austin, TX — June 2026 – Present

- Worked on implementing a fixed-point iteration method to find optimal ridge regularization parameters.
- Investigated compromises and limitations of fixed-point iteration method for optimal parameters from literature.
- Improved on the existing fixed-point iteration and moved from fixed-X error by proving convergence of a contractive mapping involving the ridge regression parameters and the generalized cross-validation formula, under some assumptions.
- Developed end-to-end algorithm for finding the optimal parameter, if exists, given a maximum ridge parameter and a level set of the GCV function of the parameter.
- Coded simulations of solution in various test cases in Python to verify results empirically, given assumptions.

Meyers Lab, The University of Texas at Austin, Austin, TX — March 2025 – Present

Honors Thesis

- Created a epidemic network generation algorithm that incorporated scale-free and small-world properties to induce real-world community structure, with the `NetworkX` package.
- Established inter-network connections inversely proportional to the degree of the nodes, to induce multi-wave epidemics by disconnecting popular nodes amongst communities.
- Implemented a SEIRS simulation approach with COVID-like parameters to evaluate transmission of the virus over time and across various communities with the `Epidemics on Networks` package.
- Disabled and suppressed types of connections to incorporate non-pharmaceutical interventions (NPIs) under thresholds.

Forecast Evaluation Project

- Evaluated local vs. aggregate infectious disease forecasts using probabilistic scoring methods, including the weighted interval score (WIS) and prediction interval coverage.
- Implemented forecast evaluation pipelines in R using the `scoringutils` package to compute and summarize WIS, including decomposition into overprediction, underprediction, and dispersion components.
- Utilized the scaled relative skill scores (SRSS) method in `scoringutils` to investigate performance differences relative to shared targets and a given model baseline.
- Developed comparative analyses using relative WIS and SRSS to assess performance differences between local forecasts and per-capita or directly scaled aggregate baselines.
- Investigated associations between forecast accuracy and locality characteristics such as population size, density, and urbanization across multiple geographic strata and forecast horizons.
- Wrote a forecasting hub study designed to evaluate sub-state infectious disease forecasting methods in the U.S. respiratory virus season.

Similarity Metrics Project

- Investigated whether local-level infectious disease forecasting provides meaningful improvements over state-level trends by quantifying divergence in epidemic curves for influenza, COVID-19, and RSV.
- Applied curve-based similarity metrics, including Discrete Fréchet distance and Dynamic Time Warping (DTW), to distinguish structural versus temporal differences between forecast trajectories.
- Plotted and analyzed similarity metrics over time and across geographic areas, finding patterns amongst geographic movement and time.

Emory Computational Mathematics for Data Science REU, Atlanta, GA — June 2025 – August 2026

- Worked on the Hybrid Regularization for Random Feature Models project under Dr. Drum.

- Investigated hybrid regularization methods such as LSQR for solving ill-posed inverse problems, with a combination of Krylov subspaces and Tikhonov regularization.
- Implemented Hybrid LSLU as an improvement over Hybrid LSQR in Random Feature Models in MATLAB with `IRTools` and evaluated methods on MNIST, PneumoniaMNIST, and OrganMNIST3D datasets.
- Analyzed instability and ill-conditioned problems when projecting into higher dimensions with Picard plots and the Discrete Picard condition.
- Achieved Hybrid LSLU performance comparable to Hybrid LSQR while significantly reducing computational cost and runtime across a variety of dimensions.

Department of Mathematics, University of North Texas, Denton, TX March 2023 – May 2024

- Conducted traffic flow research under Dr. Liu in the UNT Mathematics Department.
- Generated realistic traffic networks using `NetworkX` and modeled system dynamics using optimization techniques.
- Approximated gradient minima using Lagrange multiplier methods and solved with `NumPy`.
- Developed algorithms that reduced traffic irregularity by 31.3%.

Mixer Lab, University of North Texas, Denton, TX October 2022 – May 2024

- Conducted neuroscience research under Dr. McMahan in the UNT Learning Technologies Department.
- Developed an educational game using Unity for experimental learning studies.
- Designed recursive experimental scenarios and supported EEG-based testing.
- Worked on tools aimed at helping educators tailor instruction to individual learners.

Publications

McMahan, Timothy, Chotepong Victor Tangton, **Advay Vyas**, Chloe Huang, and Aron Yewoor (2024). "Bridging the Gap Between Perceived and Actual Personalities in Real and Virtual Worlds Through EEG Analysis". In: *Annual Review of Cybertherapy And Telemedicine 2024*, pp. 206–212. ISSN: 1554-8716.

Vyas, **Advay**, Ethan Rebello, and Jianguo Liu (2023). "A Lagrangian Approach to Loss Function Optimization on Traffic Network Regularity". In: *2023 IEEE MIT Undergraduate Research Technology Conference (URTC)*, pp. 1–5. DOI: 10.1109/URTC60662.2023.10534993.

Selected Presentations

- "Martingales and Stopping Times," Math DRP Symposium, Spring 2026
- "Quantitative Metrics for Decomposing Timing and Magnitude Differences in Influenza Forecasts," UT Austin Technology & Science Undergraduate Research Forum, Spring 2026
- "Inner Product Free Hybrid Regularization for Random Feature Models," Joint Mathematics Meetings, Spring 2026
- "Nowhere Differentiable, Everywhere Useful," Math DRP Symposium, Fall 2025
- "Hybrid Regularization for Random Feature Models," Emory REU Poster Session, Summer 2025
- "Topology and Data in Biology," Math DRP Symposium, Fall 2024
- "A Lagrangian Approach to Loss Function Optimization on Traffic Network Regularity," MIT IEEE Undergraduate Research Technology Conference, Fall 2023

Awards & Achievements

- College of Natural Sciences Advanced Summer Research Fellowship, UT Austin, 2026
- Best Use of Statistical Analysis, American Statistical Association DataFest at SMU, 2026
- Score of 3, William Lowell Putnam Mathematical Competition, 2025
- College of Natural Sciences Honors Summer Scholarship, UT Austin, 2025
- National Merit Finalist Scholarship, National Merit Scholarship Foundation, 2024
- Polymathic Scholar, UT Austin, 2024
- Undergraduate Research Fellow Scholarship, University of North Texas, 2023